Reg No.

SATHYABAMA INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)

CONTINUOUS ASSESSMENT EXAM - I

Programme: B.E – ECE -DS
Course Code: S11BLH12
Course Name: Problem Solving Technique Using C
Batch: 2025-2029

Max. Marks: 50
Duration: 2 Hours
Date : 20-02-2026
Sem : II

Answer All (5 × 10 = 50)

Q. No.	Questions	CO
	Answer all questions (2 marks each)	CO 1
1.	a. A payroll management system uses a separate block of code to calculate an employee's salary based on basic pay, allowances, and deductions. This block is reused for every employee. Which control concept is most appropriate for this purpose? i) Sequence ii) Selection iii) Iteration iv) Function	
	b. A water quality monitoring system checks contamination levels every hour and triggers an alert when the value crosses a threshold. Which control structure is most appropriate for this repeated checking? i) Sequence ii) Selection iii) Iteration iv) Function	
	c. In a solar power billing system, the system reads daily units consumed, calculates total cost, and prints the final bill. Which stage of the IPO model does unit cost × units consumed represent? i) Input ii) Process iii) Output iv) Storage	
	d. Which pseudocode keyword combination is essential for implementing multi-condition decision making in a climate alert system? i) BEGIN – END ii) INPUT – OUTPUT iii) IF – ELSE – ENDIF iv) WHILE – ENDWHILE	
	e. Why is pseudocode preferred during the design phase of sustainable software systems? i) It is compiler-dependent ii) It is visual iii) It is language-independent and modifiable iv) It reduces execution time	
	or	

	Answer all questions (2 marks each)	CO 1
2	a. A flowchart calculates the average temperature from values recorded over several days and then decides whether to display *"Heatwave Alert"*.Which flowchart symbol combination is mandatory to represent this decision? i) Process → Input ii) Input → Process iii) Decision → Output iv) Terminal → Connector	
	b. In a solar energy monitoring system, the pseudocode repeatedly reads daily energy units until all days are processed.Which pseudocode construct correctly represents this repeated execution? i) IF – ELSE – ENDIF ii) BEGIN – END iii) WHILE – ENDWHILE iv) INPUT – OUTPUT	
	c. A flowchart for a smart traffic system checks vehicle count every hour and displays *"Congestion"* if the average exceeds a limit.Which flowchart element ensures that the check happens multiple times? i) Terminal symbol ii) Connector iii) Looping flow line iv) Decision symbol	
	d. A flowchart calculates a student's average marks and classifies performance as *Excellent*, *Good*, or *Needs Improvement*.Which concept increases the complexity of this flowchart compared to a simple pass/fail system? i) Use of sequence ii) Use of iteration iii) Use of multi-way selection iv) Use of terminal symbols	
	e. A smart academic analytics system processes marks of multiple students, computes individual averages, categorizes performance into Distinction, First Class, Second Class, or Fail, and generates a summary report at the end.Which combination of flowchart concepts is essential to correctly model this system? i) Sequence + Single selection ii) Iteration + Two-way selection iii) Iteration + Multi-way selection + Sequence iv) Function call + Terminal symbols	

3.	A smart water usage monitoring system records the daily water consumption of a household for *n* days, calculates the total and	CO 1

	average water usage, and displays an "Overuse Alert" if the average consumption exceeds 150 liters per day; otherwise, it displays "Normal Usage." Write Algorithm, Flowchart and Pseudo code for this problem	& CO 2
	or	
4	A community health screening system records the body temperature of *n* patients and must identify the highest temperature recorded during the screening; if the highest temperature exceeds 39°C, the system should display "Emergency Case Detected", otherwise display "No Emergency". Write Algorithm, Flowchart and Pseudocode for this problem.	CO 1 & CO 2

5.	An online examination system records marks for *n* subjects, counts the number of failed subjects (marks < 40), and displays "Fail" with the fail count if any subject is failed; otherwise, it displays "Pass" with the total marks. Write Algorithm, Flowchart and Pseudocode for this problem.	CO 1 & CO 2
	or	
6	A smart traffic control system records vehicle count for *n* hours, counts the number of hours exceeding a given limit, and displays "Signal Optimization Required" if more than two hours cross the limit; otherwise, it displays "Traffic Under Control." Write Algorithm, Flowchart and c program for this problem.	CO 2 & CO 6
7.	An employment monitoring system records male and female recruitment counts for *n* drives and checks gender balance. If female recruits are fewer than male recruits in more than half of the drives, display "Gender Balance Not Achieved"; otherwise, display "Gender Balance Achieved". Write the Algorithm, draw the Flowchart, and implement the solution using a C program.	CO 2 & CO 6
	or	
8	An examination evaluation system computes the final score of a student using the weighted formula Final Score = (Internal Marks × 0.4) + (External Marks × 0.6); if the final score is greater than or equal to 50, the student is declared *Pass*, otherwise *Fail*. Develop the solution logic and implement it using a C program.	CO 2 & CO 6

9.	**Multiple choice questions(2 marks each)** a. What will be the value of x after execution? `Int x = 5 + 10 * 2 - 4 / 2;` i) 21 ii) 23 iii) 25 iv) 26 b. What will be the output of the expression?(5 > 3) && (8 < 6) i) 1 ii) 0 iii) True iv) Compilation error c. What will be printed? Int a = 4;printf("%d", a++ + ++a); i) 9 ii) 10 iii) 11 iv) Undefined behavior d. What is the final value of x? int `x = 10;x += 5 * 2;` i) 20 ii) 25 iii) 30 iv) 15 e. What will be the output? Int `a = 7, b = 5;printf("%d", (a > b) ? a - b : b - a);` i) 2 ii) 12 iii) -2 iv) 0	CO 3
	or	

10	**Multiple choice questions(2 marks each)**
	a. int a = 7, b = 3;printf("%d", a / b);
	a) 2 b) 2.33 c) 3 d) Error
	b. void main(){printf("%d", 5 + 2 * 3);} What will be printed?
	a) 21 b) 11 c) 15 d) 7
	c. int a = 3; printf("%d %d", a++, ++a);
	a) 3 4 b) 3 5 c) 4 5 d) Undefined
	d. printf("%d", 6 & 4);
	a) 0 b) 2 c) 4 d) 6
	e. int a = 4, b = 7;printf("%d", (a > b) ?a : b);
	a) 4 b) 7 c) 11 d) 0
		CO 3